# Affine Polarization of Gamma Normal Forms

## All derivative-square coefficients, polylogarithmic kernels, and Stieltjes averages

Research continuation prepared for Vladimir Reshetnikov

Mathematical derivations and computational checks with OpenAI assistance

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**Abstract.** The canonical integration-by-parts normal form of $(-1)^m\Gamma(x)(\Gamma(x)^{-1})^{(m)}$ contains a tractable infinite sector that is not apparent from its Bell-polynomial expansion. We determine every coefficient of $u_0^a u_1^k u_r^2$, simultaneously for all $r \geq 2$ and all $a, k \geq 0$. An affine perturbation turns the quadratic reduction into one symmetric symbol. Its coefficients are entire hypergeometric functions, with an elementary beta-integral representation and a two-variable generating function. A general symmetric-profile theorem extends the construction to products of centered Gamma ratios.

The same kernel gives a normalized Mellin–Hurwitz transform with only $r + 1$ endpoint pairs. Its Laurent expansion yields finite formulas for every beta-weighted generalized Stieltjes constant, with explicit complete-harmonic coefficients and anchored primitives of every order. A Lerch deformation gives finite polylogarithmic endpoint formulas. We evaluate six quartic-weight polygamma integrals and prove a Bernoulli selector basis characterized by single-zeta closure over rational functions of the complex order. Its trigamma specialization contains one odd zeta value plus an explicit rational number. All displayed integrals in these analytic applications are ordinary convergent integrals.

The normal-form decomposition, the pure-square coefficients, and the $r = 2$ family are credited to prior work. The new coefficient theorem answers a defined subcase of the predecessor's next-Gamma-projection question; it does not solve its full mixed $u_2, u_3$ sector or the general harmonic-tail rank conjecture. Proofs, 2,187 exact assertions, 56 numerical diagnostics, and additive integration instructions accompany the article. Historical priority beyond the inspected sources is not asserted.

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Source identity and inspection limits are recorded in Section 10.

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# 1 The continuation problem and the main result

The manuscript's Gamma hierarchy has two distinct kinds of questions. The first is algebraic: which differential monomials survive a specified integration-by-parts reduction, and with which rational coefficients? The second is analytic: how do their integrals, spectral derivatives, and normalized primitives reduce to special functions? Neither question should be replaced by an assertion of independence of numerical constants.

The immediate source for this continuation is *Exact Harmonic-Tail Reconstruction and Variational Gamma Identities* [2]. Its Section 4 determines the complete normal projection onto $u_0, u_1, u_2$. Its final research question 5 asks for the next projection, including $u_3^2$ with arbitrary powers of $u_1, u_2$. We solve the subfamily with no additional $u_2$ factor, and simultaneously solve its analogue for every derivative order $r \geq 2$. In particular, the result is not only a table at the next weight.

Let $\mathcal{D}u_j = u_{j+1}$ in $\mathbb{Q}[u_0, u_1, \ldots]$. Write $\mathcal{N}$ for the canonical projection onto monomials which are either powers of $u_0$ or have their largest positive index repeated at least twice. Define

$$A_0 = 1, \qquad A_{m+1} = u_0 A_m - \mathcal{D}A_m, \qquad N_m = \mathcal{N}(A_m). \tag{1.1}$$

The decomposition and all conventions are proved or specified in the next section. Under $u_j = \psi^{(j)}(x)$,

$$A_m(x) = (-1)^m \Gamma(x) \frac{\mathrm{d}^m}{\mathrm{d}x^m} \frac{1}{\Gamma(x)}.$$

**Theorem 1.1** (All derivative-square coefficients)**.** *Let $r \geq 2$ and $a, k \geq 0$, and put $m = a + 2k + 2r + 2$. Then*

$$\boxed{[u_0^a u_1^k u_r^2] N_m = \begin{cases} \dfrac{(-1)^r m!}{a!\, 2^{k+r+1} k! (r+1)! \prod_{j=0}^{r} (k+2j+1)}, & k \text{ even}, \\ 0, & k \text{ odd}. \end{cases}} \tag{1.2}$$

*Equivalently, with $y = u_1 t^2/2$,*

$$\sum_{m \geq 0} N_m \frac{t^m}{m!} = e^{t u_0} \left\{ f_0(y) + \sum_{r \geq 2} u_r^2 t^{2r+2} g_r(y) + R(t; u_1, u_2, \ldots) \right\}, \tag{1.3}$$

*where every monomial of $R$ contains at least three factors from $\{u_2, u_3, \ldots\}$, and*

$$f_0(y) = \frac{1}{2} \left( \cosh y + \frac{\sinh y}{y} - y \operatorname{Shi}(y) \right), \tag{1.4}$$

$$g_r(y) = \frac{(-1)^r}{(r+1)!} \partial_{y^2}^{r+1} \cosh y = \frac{(-1)^r}{(2r+2)!}\, {}_0F_1\left( ; r + \frac{3}{2}; \frac{y^2}{4} \right). \tag{1.5}$$

*Here $\operatorname{Shi}(y) = \int_0^y \sinh v \, \mathrm{d}v/v$, and $\partial_{y^2} = (2y)^{-1} \partial_y$ acts on even power series. All apparent singularities at $y = 0$ are removable.*

The phrase "quadratic sector" refers to the number of factors having index at least two. It does not mean the total factor degree, and it does not discard arbitrary powers of $u_0$ or $u_1$. Formula (1.3) is an identity of formal series, not an approximation to a Gamma ratio on an interval. Its proof is in Section 3.

### 1.1 The overlap and the advance

At $k = 0$, (1.2) is $(-1)^r \binom{m}{2r+2}$; the inspected predecessors already contain these pure-square coefficients. At $r = 2$ it agrees with the complete $u_2^2$ family in [2, Section 4]. Those cases are checks and inherited specializations, not additional discoveries claimed here. The advance is the single all-$r$, all-$k$ formula and its affine-symbol proof, together with the symmetric-profile and analytic consequences.

For example, the coefficients of $u_1^{2\ell} u_r^2$ at their corresponding weights $4\ell + 2r + 2$ begin as follows. These are not coefficients at one fixed weight.

| $r$ | $\ell = 0$ | $\ell = 1$ | $\ell = 2$ | $\ell = 3$ |
|---|---|---|---|---|
| 2 | 1 | 90 | 15015 | 4176900 |
| 3 | −1 | −165 | −40950 | −15261750 |
| 4 | 1 | 273 | 96390 | 47006190 |
| 5 | −1 | −420 | −203490 | −127193220 |
| 6 | 1 | 612 | 395010 | 310810500 |

Multiplication by $u_0^a$ multiplies the exponential-series coefficient by $1/a!$, as displayed in the theorem.

The analytic part is connected by the beta representation of $g_r$, not by an unproved interchange of the entire Gamma normal-form series with a singular integral. It yields finite Hurwitz and Lerch endpoint formulas, every generalized Stieltjes order, and ordinary integral identities such as

$$\int_1^3 (x-1)^2 (3-x)^2 \psi'(x)\, \mathrm{d}x = \frac{6\zeta(3)}{\pi^2}. \tag{1.6}$$

Individual polynomially weighted special-function integrals can have classical antecedents; our claim is the proved structural derivation and its explicitly scoped extension, not universal historical priority.

## 2 Canonical reduction, centering, and endpoint conventions

### 2.1 The free differential algebra

Let $\mathscr{R} = \mathbb{Q}[u_0, u_1, u_2, \ldots]$, where every polynomial uses finitely many variables. The weight of $u_j$ is $j + 1$; its factor degree is one. The derivation $\mathcal{D}$ increases weight by one and preserves factor degree. A monomial is *functional* when it is a power of $u_0$, including 1, or when its largest positive index occurs at least twice. Let $\mathscr{F}$ be the rational span of these monomials.

The following is the normal-form input used in the repository [1, 2]. The terminology and the underlying complement also occur in the classical differential-algebra construction [3, Section 4.4.1]. We include a proof to make our coefficient conventions independent of a particular implementation.

**Lemma 2.1** (The functional-monomial decomposition)**.** *Every $P \in \mathscr{R}$ has a unique decomposition*

$$P = \mathcal{N}(P) + \mathcal{D}\mathcal{T}(P), \qquad \mathcal{N}(P) \in \mathscr{F}, \qquad [1]\mathcal{T}(P) = 0. \tag{2.1}$$

*Both maps are rational linear and preserve factor degree. On positive weight $w$, $\mathcal{N}(P)$ has weight $w$ and $\mathcal{T}(P)$ has weight $w - 1$.*

*Proof.* Suppose a monomial has its largest index $r \geq 1$ exactly once. Write it as $u_r u_{r-1}^a V$, where $V$ uses indices below $r - 1$. Then

$$u_r u_{r-1}^a V = \mathcal{D}\left(\frac{u_{r-1}^{a+1} V}{a+1}\right) - \frac{u_{r-1}^{a+1} \mathcal{D} V}{a+1}. \tag{2.2}$$

Every monomial of the remainder has largest index below $r$. Repeating this replacement terminates and produces a functional remainder. Constants of primitives can be set to zero.

For uniqueness, if a nonconstant polynomial $H$ has largest variable index $j$, the terms $u_{j+1}\partial H/\partial u_j$ in $\mathcal{D}H$ are nonzero and have largest index $j+1$ exactly once. All other terms have smaller largest index. They cannot be canceled by a functional polynomial. Thus $\mathcal{D}H \in \mathscr{F}$ forces $H$ constant, proving both directness and uniqueness after the stated normalization. □

This is a vector-space decomposition. For instance, $\mathcal{N}(u_1) = 0$ but $\mathcal{N}(u_1^2) = u_1^2$; replacing $\mathcal{N}(PQ)$ by $\mathcal{N}(P)\mathcal{N}(Q)$ is incorrect. Another useful check is $\mathcal{N}(u_0u_4) = u_2^2$. Truncating away $u_4$ before reduction would lose that term. These are safeguards for the present derivation, not newly discovered errors in sources which already state the distinction.

## 2.2 The centered Gamma representative

The exponential generating function of (1.1) is

$$\mathcal{A}(t) = \sum_{m\geq 0} A_m \frac{t^m}{m!} = \exp\left(\sum_{j\geq 0} \frac{(-1)^j u_j t^{j+1}}{(j+1)!}\right). \tag{2.3}$$

It follows either by the exponential Bell formula or by differentiating with respect to $t$ and using the defining recurrence. Under the digamma substitution it is $\Gamma(x)/\Gamma(x-t)$.

Define

$$\mathcal{C}(t) = e^{t\mathcal{D}/2}\mathcal{A}(t) = \exp\left(\sum_{j\geq 0} \frac{u_{2j} t^{2j+1}}{2^{2j}(2j+1)!}\right). \tag{2.4}$$

For any formal differential series $P(t)$,

$$(e^{t\mathcal{D}/2} - 1)P(t) = \mathcal{D}\left(\sum_{n\geq 1} \frac{t^n \mathcal{D}^{n-1} P(t)}{2^n n!}\right).$$

Therefore $\mathcal{N}\mathcal{C}(t) = \mathcal{N}\mathcal{A}(t)$. Analytically the centered representative is $\Gamma(x+t/2)/\Gamma(x-t/2)$, initially near $t=0$ away from Gamma poles.

Translation $u_0 \mapsto u_0 + c$ commutes with $\mathcal{D}$ and preserves the functional space. Uniqueness in Lemma 2.1 implies that it commutes with $\mathcal{N}$. Thus $\mathcal{N}\mathcal{A}(t)$ has an Appell factor $e^{tu_0}$. The centered series also shows factor parity: each exponent term has odd weight and factor degree one, so every surviving monomial has factor degree congruent to its weight modulo two.

**Lemma 2.2** (Appell covariance of the primitive)**.** *Suppose $P(t)$ and $\mathcal{N}P(t)$ both satisfy $\partial_{u_0}P = tP$. If $P - \mathcal{N}P = \mathcal{D}T$ with $[1]T = 0$, then*

$$\partial_{u_0}T - tT = c(t)$$

*for a scalar series $c(t)$. Consequently, modulo a scalar series, $T = e^{tu_0}B(t; u_1, u_2, \ldots)$, allowing a harmless Laurent power of $t$ if necessary.*

*Proof.* The operators $\partial_{u_0}$ and $\mathcal{D}$ commute. Apply $\partial_{u_0} - t$ to $\mathcal{D}T = P - \mathcal{N}P$. Its result is zero, so the kernel statement in Lemma 2.1 makes $\partial_{u_0}T - tT$ scalar. Solving this elementary equation in $u_0$ gives $T = e^{tu_0}B - c(t)/t$. □

This small observation matters in the affine-symbol proof: after perturbation, the quadratic part of the primitive has the same exponential weight as the density and coefficients independent of the affine coordinate. We do not enlarge the quotient by arbitrary $x$-dependent primitives.

### 2.3 What an analytic substitution does and does not imply

For every fixed $m$, let $T_m = \mathcal{T}(A_m)$. On an interval $[\alpha, \beta] \subset (0, \infty)$,

$$\int_\alpha^\beta (-1)^m \Gamma(x)(\Gamma(x)^{-1})^{(m)}\,\mathrm{d}x = T_m(\psi(\beta), \ldots) - T_m(\psi(\alpha), \ldots) + \int_\alpha^\beta N_m(\psi(x), \ldots)\,\mathrm{d}x. \tag{2.5}$$

If a fixed-coordinate finite part at zero is used instead, the endpoint is $T_m(\psi(1), \ldots) - [x^0]T_m(\psi(x), \ldots)$, not zero. This inherited endpoint convention [1] is retained.

At positive integral endpoints the entries are generalized harmonic numbers:

$$\psi(N+1) = H_N - \gamma, \qquad \psi^{(j)}(N+1) = (-1)^{j+1} j!(\zeta(j+1) - H_N^{(j+1)}) \quad (j \geq 1). \tag{2.6}$$

These follow from the standard polygamma recurrence [6]. Thus the exact normal-form certificates also give finite identities between Gamma derivatives and harmonic-tail polynomials. They do not by themselves evaluate every integral of a remaining normal monomial.

## 3 Affine polarization and the complete square sector

### 3.1 A constant-coefficient quadratic symbol

Let $p, q, t$ be formal parameters and make the differential substitution

$$u_0 = q + px + \varepsilon v(x), \qquad u_1 = p + \varepsilon v'(x), \qquad u_j = \varepsilon v^{(j)}(x) \quad (j \geq 2). \tag{3.1}$$

Put $W = e^{t(q+px)}$ and $\kappa = tp$, so $W' = \kappa W$. For a quadratic differential polynomial in $v$ with constant coefficients, polarization introduces two derivative variables $\xi_1, \xi_2$. Modulo a total derivative with weight $W$, the relation is

$$\xi_1 + \xi_2 + \kappa = 0.$$

Set $\xi_1 = \xi$, $\xi_2 = -\xi - \kappa$, and

$$\eta = -\xi(\xi + \kappa). \tag{3.2}$$

The symbol of $(v^{(r)})^2$ is $\eta^r$, while the symbol of $vv'$ is $-\kappa/2$. A weighted derivative is annihilated because its symbol is multiplied by $\xi_1 + \xi_2 + \kappa$.

**Lemma 3.1** (Uniqueness of the square symbol)**.** *If a finite sum $W \sum_{r=0}^{M} c_r (v^{(r)})^2$ is a weighted total derivative of a constant-coefficient quadratic differential expression, then every $c_r$ is zero. The same conclusion holds coefficientwise for formal series in $t$.*

*Proof.* Its symbol is $\sum c_r \eta^r = 0$. Since $\eta$ is a polynomial of degree two in $\xi$, distinct powers have different degrees and are linearly independent. The formal statement follows at every finite coefficient of $t$. Conversely, the symmetric symbols invariant under $\xi \mapsto -\xi - \kappa$ form the polynomial ring in $\eta$, so this symbol also supplies the usual diagonal reduction of quadratic forms. □

Lemma 2.2 ensures that this is exactly the quotient needed after (3.1): the quadratic part of the primitive of $\mathcal{C} - \mathcal{N}\mathcal{C}$ is $W$ times an expression whose coefficients do not depend on $x$. An arbitrary $x$-dependent antiderivative, which would invalidate the symbol argument, is not being allowed.

### 3.2 The symbol of the centered Gamma exponential

Define the even differential series

$$K_t(z) = \frac{2\sinh(tz/2)}{z}, \qquad K_t(0) = t.$$

The exponent in (2.4) is $K_t(\mathcal{D})u_0$. Its affine part is $t(q+px)$, so the quadratic perturbation of $\mathcal{C}$ is

$$[\varepsilon^2]\mathcal{C} = \frac{W}{2}\big(K_t(\mathcal{D})v\big)^2. \tag{3.3}$$

Now set $y = t^2p/2$ and $h = t(\xi+\kappa/2)$; then $h^2 = y^2 - t^2\eta$. The product-to-sum identity for hyperbolic sines gives the exact symbol

$$\begin{aligned}\frac{1}{2}K_t(-\xi-\kappa)K_t(\xi) &= \frac{\cosh y - \cosh\sqrt{y^2-t^2\eta}}{\eta} \\ &= \sum_{r\ge 0}\frac{(-1)^r t^{2r+2}}{(r+1)!}\partial_{y^2}^{r+1}\cosh y\,\eta^r.\end{aligned} \tag{3.4}$$

The square root introduces no branch: $\cosh\sqrt{z}$ means its entire series $\sum_{n\ge0} z^n/(2n)!$. The numerator is divisible by $\eta$.

On the other hand, normal support and the Appell factor imply (1.3), initially with unknown $f_0, g_r$. No normal monomial has exactly one factor of index at least two. If it has exactly two, their indices must coincide. All other such monomials contribute only at order $\varepsilon^3$ or higher. The quadratic perturbation of its low-index part is

$$W\left\{\frac{t^2 f_0}{2}v^2 + \frac{t^3 f_0'}{2}vv' + \frac{t^4 f_0''}{8}(v')^2\right\}.$$

After replacing the mixed term by its symbol, the coefficients of $\eta^0$ and $\eta^1$ are

$$\frac{t^2}{2}(f_0 - yf_0'), \qquad \frac{t^4}{8}f_0''.$$

For $r \ge 2$ the coefficient of $\eta^r$ is simply $t^{2r+2}g_r(y)$. Lemma 3.1 and (3.4) therefore prove the derivative formula for $g_r$ in Theorem 1.1.

The coefficient of $\eta^0$ gives

$$f_0 - yf_0' = \frac{\sinh y}{y}. \tag{3.5}$$

The normal form excludes a term linear in $u_1$, hence a term linear in $y$. With $f_0(0) = 1$, the unique solution is (1.4). The $\eta^1$ equation agrees with its second derivative and supplies an independent normalization check.

### 3.3 The beta representation and explicit coefficients

**Proposition 3.2** (Entire kernels and their recurrences)**.** *For every $r \ge 0$ define $g_r(y) = (-1)^r\partial_{y^2}^{r+1}\cosh y/(r+1)!$. Then*

$$g_r(y) = c_r\int_0^1 (1-v^2)^r\cosh(yv)\,\mathrm{d}v, \qquad c_r = \frac{(-1)^r}{2^{2r+1}r!(r+1)!}. \tag{3.6}$$

*They are even entire functions and satisfy*

$$\sum_{r\ge 0} g_r(y)w^r = \frac{\cosh y - \cosh\sqrt{y^2-w}}{w}, \tag{3.7}$$

$$g_{r+1}(y) = -\frac{1}{r+2}\partial_{y^2} g_r(y), \tag{3.8}$$

$$yg_r'' + (2r+2)g_r' - yg_r = 0, \qquad g_r(0) = \frac{(-1)^r}{(2r+2)!}. \tag{3.9}$$

*In* (3.7)*, the terms* $r = 0, 1$ *are symbol coefficients; they are not* $f_0$ *and a nonexistent normal* $u_1$ *term.*

*Proof.* Expanding the hyperbolic cosine in (3.6) is uniformly convergent on compact sets. For every even $k \ge 0$,

$$\int_0^1 v^k(1-v^2)^r\,\mathrm{d}v = \frac{1}{2}B\left(\frac{k+1}{2}, r+1\right) = \frac{2^r r!}{\prod_{j=0}^r (k+2j+1)}.$$

Thus

$$k![y^k]g_r(y) = \frac{(-1)^r}{2^{r+1}(r+1)!\prod_{j=0}^r(k+2j+1)} \quad (k \text{ even}), \tag{3.10}$$

and odd coefficients vanish. Comparing with the differentiated cosine series proves the beta identity and the ${}_0F_1$ expression. Taylor expansion in $w$ proves (3.7); differentiation proves (3.8). Substituting the coefficients proves (3.9). Finally, expand $e^{tu_0}$ and set $y = u_1t^2/2$ to obtain (1.2). □

The first two high-index kernels are elementary:

$$g_2(y) = \frac{(y^2+3)\sinh y - 3y\cosh y}{48y^5}, \tag{3.11}$$

$$g_3(y) = -\frac{(y^3+15y)\cosh y - (6y^2+15)\sinh y}{384y^7}. \tag{3.12}$$

At the origin these formulas should be evaluated by their even Taylor series or by ${}_0F_1$, rather than by subtracting nearly equal floating numbers. The recurrence generates every higher elementary hyperbolic expression, while the beta form is preferable for the analytic bridge.

## 4 Symmetric profiles and mixed Gamma ratios

The affine argument does not depend on the particular uniform interval average in the Gamma ratio. Let $K(z) \in \mathbb{C}[[z^2]]$ be even, with $K(0) = 1$, and form

$$\mathcal{C}_K(t) = \exp\big(tK(t\mathcal{D})u_0\big). \tag{4.1}$$

Its coefficients are differential polynomials, and its normal form again has an Appell factor. Denote the coefficient of $u_r^2t^{2r+2}$ after removing that factor by $g_{K,r}(y)$, with $y = u_1t^2/2$.

**Theorem 4.1** (The symmetric-profile coefficient theorem)**.** *For every* $r \ge 2$*,*

$$\boxed{g_{K,r}(y) = [w^r]\frac{1}{2}K\left(\sqrt{y^2-w}-y\right)K\left(\sqrt{y^2-w}+y\right).} \tag{4.2}$$

*The product is even in the square-root variable, so its interpretation as a formal series in* $y, w$ *is unambiguous. Formula* (4.2) *determines all coefficients of* $u_0^a u_1^k u_r^2$ *in* $\mathcal{NC}_K$*.*

*Proof.* Under (3.1), the quadratic term is $t^2W(K(t\mathcal{D})v)^2/2$. Its symbol is $t^2K(h-y)K(h+y)/2$, where $h^2=y^2-t^2\eta$. The primitive has Appell covariance by Lemma 2.2, and normal support excludes any other contribution to $\eta^r$ for $r\geq 2$. Comparing coefficients of $t^{2r+2}\eta^r$ proves the formula. Evenness of $K$ makes the product invariant under $h\mapsto -h$, justifying its formal meaning. □

### 4.1 A finite collection of Gamma scales

For nonzero scales $\rho_1,\dots,\rho_d$ and constants $c_i$ with $\sum c_i=1$, put

$$K(z)=\sum_{i=1}^{d} c_iK_{\rho_i}(z), \qquad K_\rho(z)=\frac{2\sinh(\rho z/2)}{\rho z}.$$

Under the digamma substitution, (4.1) becomes

$$\prod_{i=1}^{d}\left(\frac{\Gamma(x+\rho_it/2)}{\Gamma(x-\rho_it/2)}\right)^{c_i/\rho_i}. \tag{4.3}$$

The powers are defined by the analytic logarithm which vanishes at $t=0$. For positive real scales, real $x>0$, and sufficiently small real $t$, this is the usual positive real power.

For compactness write $D_n=\partial_{y^2}^n$, $A_{ij}(y)=\cosh((\rho_i-\rho_j)y/2)$, and $B_{ij}(y)=\cosh((\rho_i+\rho_j)y/2)$. Theorem 4.1 gives the explicit mixed-scale formula

$$g_{K,r}(y)=\frac{(-1)^r}{(r+1)!}\Bigg\{\sum_i\frac{c_i^2}{\rho_i^2}D_{r+1}\cosh(\rho_iy) + 2\sum_{i<j}\frac{c_ic_j}{\rho_i\rho_j}\left(A_{ij}D_{r+1}B_{ij}-B_{ij}D_{r+1}A_{ij}\right)\Bigg\}. \tag{4.4}$$

Indeed, expand the product in (4.2) and use the hyperbolic product-to-sum identities before Taylor expanding in $w$. Coincident scales have a regular limit; no division by $\rho_i-\rho_j$ appears.

As one concrete check, take $c_1=c_2=1/2$, $\rho_1=1$, and $\rho_2=2$. The coefficient of $u_2^2$ in the weight-six exponential coefficient, normalized by 6!, is

$$[u_2^2]\,6![t^6]\mathcal{NC}_K(t)=\frac{227}{32}. \tag{4.5}$$

It can also be verified without the symbol: the exponent is $tu_0+(5/48)t^3u_2+(17/3840)t^5u_4+\cdots$, and $\mathcal{N}(u_0u_4)=u_2^2$. The two contributions are $51/16$ and $125/32$.

The general theorem is useful even when $K$ is a polynomial rather than a Gamma profile. Its proof does not use analytic convergence of $K$. For convergent profiles the resulting kernels may have additional analytic representations; the remainder of this article treats the uniform Gamma profile, whose beta weight gives particularly short endpoint formulas.

## 5 A finite-endpoint Mellin–Hurwitz transform

Fix $r\geq 2$ and initially take real $a>b>0$. Define

$$\mathcal{H}_r(s;a,b)=\frac{1}{\Gamma(s)}\int_0^\infty\frac{t^{s-1}e^{-at}g_r(bt)}{1-e^{-t}}\,\mathrm{d}t, \qquad \Re s>1. \tag{5.1}$$

At zero the integrand is $O(t^{\Re s-2})$, and at infinity the exponential decay is governed by $a-b>0$. These conditions justify ordinary integration and all fixed-order spectral derivatives in the stated half-plane.

The Hurwitz zeta convention is $\zeta(s,x)=\sum_{n\ge0}(n+x)^{-s}$ for $\Re s>1$, continued meromorphically in $s$. Its standard Mellin representation and spatial derivative are recorded in [7]. They also follow directly from the series in their common domain.

**Theorem 5.1** (The beta average and its finite endpoints)**.** *With $c_r$ as in (3.6),*

$$\mathcal{H}_r(s;a,b)=\frac{c_r}{2}\int_{-1}^{1}(1-v^2)^r\zeta(s,a+bv)\,\mathrm{d}v. \tag{5.2}$$

*This provides meromorphic continuation to every $s$, with the single possible pole $s=1$, whose residue is*

$$g_r(0)=\frac{(-1)^r}{(2r+2)!}. \tag{5.3}$$

*Define*

$$w_{rj}(b)=\frac{(-1)^j j!\binom{r}{j-r}}{2^{j+2}r!(r+1)!b^{j+1}},\qquad r\le j\le 2r. \tag{5.4}$$

*Then, as an identity of meromorphic functions,*

$$\boxed{\mathcal{H}_r(s;a,b)=\sum_{j=r}^{2r}w_{rj}(b)\frac{\zeta(s-j-1,a+b)-(-1)^j\zeta(s-j-1,a-b)}{(1-s)_{j+1}}.} \tag{5.5}$$

*At apparent poles of the separate summands, the entire sum is to be continued jointly. No pole other than $s=1$ survives.*

*Proof.* Insert (3.6) into (5.1). Absolute convergence permits interchanging the integrals. Combining the two exponentials in the hyperbolic cosine yields (5.2). On every compact subset of $\Re x>0$, $\zeta(s,x)-1/(s-1)$ is holomorphic and locally bounded in $s$. Its compact $v$-integral is therefore entire, proving the continuation and the residue. The integral of the weight is the one already used in (3.10).

For the endpoint identity let $P(v)=(1-v^2)^r$. Repeated spatial antidifferentiation gives

$$\frac{\mathrm{d}^{j+1}}{\mathrm{d}v^{j+1}}\frac{\zeta(s-j-1,a+bv)}{b^{j+1}(1-s)_{j+1}}=\zeta(s,a+bv),$$

initially for $s$ outside the finite exceptional set. Integrating by parts until the polynomial vanishes leaves

$$\sum_{j=0}^{2r}\frac{(-1)^j}{b^{j+1}(1-s)_{j+1}}[P^{(j)}(v)\zeta(s-j-1,a+bv)]_{v=-1}^{v=1}.$$

The derivatives with $j<r$ vanish at both endpoints, and for $r\le j\le 2r$,

$$P^{(j)}(1)=(-1)^r j!\binom{r}{j-r}2^{2r-j},\qquad P^{(j)}(-1)=(-1)^jP^{(j)}(1). \tag{5.6}$$

Multiplication by $c_r/2$ proves (5.5). Analytic continuation establishes it at every remaining regular point. □

The formulas extend holomorphically in $a, b$ when $\Re a > |b|$, using the right-half-plane Hurwitz branch. They are even in $b$. At $b = 0$ the compact formula gives the confluent value

$$\mathcal{H}_r(s; a, 0) = g_r(0)\zeta(s, a). \tag{5.7}$$

The negative powers of $b$ in (5.5) must cancel; one should not substitute $b = 0$ separately in those terms.

### 5.1 All logarithmically weighted Mellin integrals

Let $\mathsf{Y}_m$ be the complete exponential Bell polynomial, defined by $\exp(\sum_{j\geq 1} x_j z^j/j!) = \sum_{m\geq 0} \mathsf{Y}_m(x_1, \ldots, x_m) z^m/m!$. Differentiating the normalized Mellin factor yields, for $\Re s > 1$,

$$\partial_s^m \mathcal{H}_r(s; a, b) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1} e^{-at} g_r(bt)}{1 - e^{-t}} \times \mathsf{Y}_m\big(\log t - \psi(s), -\psi'(s), \ldots, -\psi^{(m-1)}(s)\big)\, \mathrm{d}t. \tag{5.8}$$

For $m = 0$ the Bell factor is one. For $m = 1, 2$ the factors are $\log t - \psi(s)$ and $(\log t - \psi(s))^2 - \psi'(s)$. This is an exact normalization, not the unnormalized rule of merely inserting $(\log t)^m$. Endpoint differentiation of (5.5) therefore gives finite evaluations of all these convergent logarithmic integrals.

## 6 Every Stieltjes order and anchored parameter primitives

We use the Laurent convention

$$\zeta(1 + \varepsilon, x) = \frac{1}{\varepsilon} + \sum_{m\geq 0} \frac{(-1)^m}{m!} \gamma_m(x) \varepsilon^m, \qquad \gamma_0(x) = -\psi(x). \tag{6.1}$$

Define the ordinary weighted averages

$$\mathcal{B}_{r,m}(a, b) = \frac{c_r}{2} \int_{-1}^{1} (1 - v^2)^r \gamma_m(a + bv)\, \mathrm{d}v. \tag{6.2}$$

Consequently

$$\mathcal{H}_r(1 + \varepsilon; a, b) = \frac{g_r(0)}{\varepsilon} + \sum_{m\geq 0} \frac{(-1)^m}{m!} \mathcal{B}_{r,m}(a, b) \varepsilon^m.$$

This expansion is locally uniform for the spatial parameters under the conditions of Theorem 5.1.

### 6.1 Finite complete-harmonic coefficients

For $j \geq 1$, let

$$E_j(\varepsilon) = \prod_{n=1}^{j} (1 - \varepsilon/n)^{-1} = \sum_{\ell\geq 0} h_\ell^{(j)} \varepsilon^\ell. \tag{6.3}$$

These are complete homogeneous polynomials in $1, 1/2, \ldots, 1/j$. They can be calculated from generalized harmonic numbers without expanding a product:

$$h_0^{(j)} = 1, \qquad \ell h_\ell^{(j)} = \sum_{d=1}^{\ell} H_j^{(d)} h_{\ell-d}^{(j)}. \tag{6.4}$$

For example, $h_1^{(j)} = H_j$ and $h_2^{(j)} = ((H_j)^2 + H_j^{(2)})/2$.

**Theorem 6.1** (All-order Stieltjes endpoint formula)**.** *Let $D_{rj} = w_{rj}(b)/j!$, and use $\zeta^{(k)}$ for a derivative in the spectral variable. For every $m \geq 0$,*

$$\boxed{\mathcal{B}_{r,m}(a,b) = (-1)^{m+1} m! \sum_{j=r}^{2r} D_{rj} \sum_{k=0}^{m+1} \frac{h_{m+1-k}^{(j)}}{k!} \left\{ \zeta^{(k)}(-j, a+b) - (-1)^j \zeta^{(k)}(-j, a-b) \right\}.} \tag{6.5}$$

*There are only $r+1$ endpoint pairs, and the largest spectral derivative order is $m+1$.*

*Proof.* At $s = 1+\varepsilon$,

$$(1-s)_{j+1} = -j!\varepsilon \prod_{n=1}^{j} (1 - \varepsilon/n).$$

Set $F_j(\varepsilon) = \zeta(-j+\varepsilon, a+b) - (-1)^j \zeta(-j+\varepsilon, a-b)$. Theorem 5.1 becomes

$$\mathcal{H}_r(1+\varepsilon; a, b) = -\frac{1}{\varepsilon} \sum_{j=r}^{2r} D_{rj} E_j(\varepsilon) F_j(\varepsilon). \tag{6.6}$$

The constant coefficient of the numerator is $-g_r(0)$. The coefficient of $\varepsilon^{m+1}$ is a finite convolution of (6.3) with the Taylor series of $F_j$. Comparison with (6.1) proves (6.5). Taking a logarithmic derivative of (6.3) proves (6.4). □

At $m = 0$ the inner expression is $F_j'(0) + H_j F_j(0)$. At $m = 1$ it is $F_j''(0)/2 + H_j F_j'(0) + ((H_j)^2 + H_j^{(2)}) F_j(0)/2$. Thus the harmonic counterterms are required even though every spatial integral in (6.2) is nonsingular. They arise from the finite endpoint denominators, not from an arbitrary regularization.

### 6.2 Spatial derivatives

Differentiation of the compact formula gives, for $L \geq 0$,

$$\partial_a^L \mathcal{H}_r(s; a, b) = (-1)^L (s)_L \mathcal{H}_r(s+L; a, b). \tag{6.7}$$

For $L \geq 1$, comparison at $s = 1+\varepsilon$ yields

$$\partial_a^L \mathcal{B}_{r,m}(a,b) = (-1)^{m+L} m! [\varepsilon^m] (1+\varepsilon)_L \mathcal{H}_r(1+L+\varepsilon; a, b). \tag{6.8}$$

In particular,

$$\partial_a \mathcal{B}_{r,m} = (-1)^{m+1} \left\{ \partial_s^m \mathcal{H}_r(2; a, b) + m \partial_s^{m-1} \mathcal{H}_r(2; a, b) \right\},$$

where the second term is omitted for $m = 0$. The finite product $(1+\varepsilon)_L$ gives another explicit generalized-harmonic coefficient expansion when desired.

### 6.3 First primitives and the Gamma endpoint

**Corollary 6.2** (An anchored first primitive)**.** *For $a, a_0 > b > 0$ and $m \geq 0$,*

$$\int_{a_0}^{a} \mathcal{B}_{r,m}(\alpha, b)\, \mathrm{d}\alpha = \frac{(-1)^{m+1}}{m+1} \, \partial_s^{m+1} \big( \mathcal{H}_r(s; a, b) - \mathcal{H}_r(s; a_0, b) \big) \Big|_{s=0}. \tag{6.9}$$

*For $m = 0$ this is the ordinary log-Gamma identity*

$$\int_{a_0}^{a} \mathcal{B}_{r,0}(\alpha, b)\, \mathrm{d}\alpha = -\frac{c_r}{2} \int_{-1}^{1} (1-v^2)^r \log \frac{\Gamma(a+bv)}{\Gamma(a_0+bv)}\, \mathrm{d}v. \tag{6.10}$$

*Proof.* Equation (6.7) with $L = 1$ gives $\partial_a \mathcal{H}_r(s) = -s\mathcal{H}_r(1+s)$. Its coefficient of $s^{m+1}$ is $(-1)^{m+1}\mathcal{B}_{r,m}/m!$, proving (6.9) after differentiation and anchoring. Use $\zeta'(0,x) = \log\Gamma(x) - \frac{1}{2}\log(2\pi)$ [7] in (5.2); the constant cancels between the two endpoints, giving (6.10). □

### 6.4 An explicit primitive of every depth

The same argument can be made completely normalized at arbitrary integration depth, without introducing unspecified constants. For $L \geq 1$ set

$$U_L(s;a,b) = \frac{\mathcal{H}_r(s-L;a,b)}{(1-s)_L}.$$

Define

$$V_L(s;a,a_0,b) = U_L(s;a,b) - \sum_{j=0}^{L-1} \frac{(a-a_0)^j}{j!}\partial_a^j U_L(s;a_0,b) - \frac{g_r(0)(a-a_0)^L}{L!(s-1)}. \tag{6.11}$$

**Theorem 6.3** (All anchored Stieltjes primitives)**.** *The function $V_L$ is holomorphic near $s = 1$, and*

$$J_{L,m}(a;a_0,b) = (-1)^m m![\varepsilon^m]V_L(1+\varepsilon;a,a_0,b) \tag{6.12}$$

*satisfies $\partial_a^L J_{L,m} = \mathcal{B}_{r,m}(a,b)$ and $\partial_a^j J_{L,m}(a_0;a_0,b) = 0$ for $0 \leq j < L$. It is the unique $L$-fold primitive with those anchor conditions.*

*Proof.* The identity $\partial_a^L U_L = \mathcal{H}_r(s;a,b)$ follows from (6.7). The residue of $U_L$ at $s = 1$ is a polynomial of degree at most $L$ in $a$, because $\zeta(1-L,a+bv) = -B_L(a+bv)/L$. Its $L$th derivative is $g_r(0)$. Taylor subtraction through degree $L-1$ therefore leaves exactly the last pole written in (6.11); removing it proves holomorphy. Differentiating $L$ times leaves $\mathcal{H}_r(s;a,b) - g_r(0)/(s-1)$. Coefficient comparison proves the claimed derivative. All lower anchor derivatives vanish by construction. The difference of any two such primitives is a polynomial of degree below $L$ with all its anchor derivatives zero, hence vanishes. □

## 7 Lerch deformation and finite polylogarithmic closure

For $|z| < 1$, replace the denominator in (5.1) by $1 - ze^{-t}$ and define

$$\mathcal{H}_r(s;a,b;z) = \frac{1}{\Gamma(s)}\int_0^\infty \frac{t^{s-1}e^{-at}g_r(bt)}{1-ze^{-t}}\,\mathrm{d}t, \qquad \Re s > 0. \tag{7.1}$$

Let $\Phi(z,s,x) = \sum_{n\geq 0} z^n(n+x)^{-s}$ in the right half-plane. The Lerch series, its Mellin formula, and its relation to polylogarithms are classical [9]. Termwise spatial differentiation gives $\partial_x\Phi(z,s,x) = -s\Phi(z,s+1,x)$.

**Theorem 7.1** (Lerch and polylogarithmic endpoints)**.** *For $|z| < 1$,*

$$\mathcal{H}_r(s;a,b;z) = \frac{c_r}{2}\int_{-1}^1 (1-v^2)^r \Phi(z,s,a+bv)\,\mathrm{d}v, \tag{7.2}$$

$$= \sum_{j=r}^{2r} w_{rj}(b)\frac{\Phi(z,s-j-1,a+b) - (-1)^j\Phi(z,s-j-1,a-b)}{(1-s)_{j+1}}. \tag{7.3}$$

*The function is entire in $s$. Hence every apparent integer pole on the right is removable, including $s = 1$. If $a \pm b$ are positive integers, every endpoint is a finite polylogarithmic expression, using*

$$\Phi(z, s, N) = z^{-N} \left( \mathrm{Li}_s(z) - \sum_{n=1}^{N-1} \frac{z^n}{n^s} \right) \quad (N \geq 1, z \neq 0). \tag{7.4}$$

*All values at $z = 0$ are supplied by analytic limits.*

*Proof.* The beta representation and absolute convergence prove (7.2). The endpoint argument in Theorem 5.1 applies unchanged. Exponential decay in $n$ makes the Lerch series locally uniformly convergent for every $s$, including all spectral derivatives. This proves entire continuation and removability of the displayed poles. Shifting the summation index proves (7.4). □

The limit $z \to 1$ is initially justified for $\Re s > 1$, where it recovers the Hurwitz transform. Continuing in $s$ before taking that limit need not be interchangeable at $s = 1$: the Hurwitz pole must be kept. At a removable integer in (7.3), the value may involve *order derivatives* of polylogarithms; a formula containing a vanishing denominator should not be read as an ordinary substitution rule.

### 7.1 Rational spatial endpoints

For $0 < z < 1$, let $w = z^{1/q}$, $\omega = e^{2\pi \mathrm{i}/q}$, and take positive integers $A, q$. The roots-of-unity filter in the absolutely convergent polylogarithm series gives

$$\begin{aligned} \Phi(z, s, A/q) = q^{s-1} w^{-A} \sum_{\ell=0}^{q-1} \omega^{-\ell A} \, \mathrm{Li}_s(w\omega^\ell) \\ - q^s w^{-A} \sum_{\substack{1 \leq n < A \\ n \equiv A \pmod{q}}} \frac{w^n}{n^s}. \end{aligned} \tag{7.5}$$

Indeed the finite Fourier sum retains exactly the positive integers congruent to $A$ modulo $q$, and the finite prefix removes those below $A$. The formula holds for every complex $s$ by normal convergence. Its order derivatives give finite log-weighted polylogarithm reductions for all spectral derivatives of the rational-endpoint transform.

### 7.2 An explicit expression with only three polylogarithms

Put

$$M(t) = (t^2 + 3) \sinh t - 3t \cosh t,$$

and, for $z \neq 0$,

$$J_\sigma^\pm(z) = (z^{-3} \pm z^{-1}) \, \mathrm{Li}_\sigma(z) - z^{-2} - 2^{-\sigma} z^{-1}.$$

For $r = 2, a = 2, b = 1$, (7.3) reads

$$\mathcal{H}_2(s; 2, 1; z) = \frac{J_{s-3}^-(z)}{96(1-s)_3} - \frac{J_{s-4}^+(z)}{32(1-s)_4} + \frac{J_{s-5}^-(z)}{32(1-s)_5}. \tag{7.6}$$

At $s = 6$ there is no exceptional denominator. Since $g_2(t) = M(t)/(48t^5)$, this gives the ordinary convergent identity

$$\int_0^\infty \frac{e^{-2t}M(t)}{1-ze^{-t}}\,\mathrm{d}t = \frac{4}{z^2} + \frac{5}{4z} - \frac{1-z^2}{z^3}\left(\mathrm{Li}_3(z) + \frac{3}{2}\,\mathrm{Li}_1(z)\right) - \frac{3(1+z^2)}{2z^3}\,\mathrm{Li}_2(z), \qquad 0 < z < 1. \tag{7.7}$$

The finite limit at $z = 1$ is $21/4 - \pi^2/2$. The individual negative powers of $z$ cancel at $z = 0$. These endpoint cancellations are most transparent in the integral or the Lerch series, rather than in the expanded polylogarithmic expression.

## 8 Quartic weights: polygamma, Stieltjes, and hyperbolic integrals

Take $r = 2, a = 2, b = 1$ and define $W(x) = (x-1)^2(3-x)^2$. Then

$$\mathcal{H}_2(s;2,1) = \frac{1}{768}\int_1^3 W(x)\zeta(s,x)\,\mathrm{d}x. \tag{8.1}$$

Integer translation of the Hurwitz zeta in (5.5) gives

$$\mathcal{H}_2(s;2,1) = -\frac{1+2^{3-s}}{96(1-s)_3} - \frac{2\zeta(s-4)-1-2^{4-s}}{32(1-s)_4} - \frac{1+2^{5-s}}{32(1-s)_5}. \tag{8.2}$$

This is a three-term meromorphic identity; the right side is not to be evaluated term by term at its apparent poles.

**Proposition 8.1** (Six ordinary polygamma integrals)**.** *The following identities hold:*

$$\int_1^3 W(x)\psi(x)\,\mathrm{d}x = \frac{83}{180} - 8\zeta'(-3), \tag{8.3}$$

$$\int_1^3 W(x)\psi'(x)\,\mathrm{d}x = \frac{6\zeta(3)}{\pi^2}, \tag{8.4}$$

$$\int_1^3 W(x)\psi''(x)\,\mathrm{d}x = -48\zeta'(-1) + 8\log 2 - 14, \tag{8.5}$$

$$\int_1^3 W(x)\psi^{(3)}(x)\,\mathrm{d}x = 24\log(4\pi) - 60, \tag{8.6}$$

$$\int_1^3 W(x)\psi^{(4)}(x)\,\mathrm{d}x = 48\gamma + 24\log 2 - 46, \tag{8.7}$$

$$\int_1^3 W(x)\psi^{(5)}(x)\,\mathrm{d}x = 84 - 8\pi^2. \tag{8.8}$$

*Proof.* Take the Laurent coefficient at $s = 1$ and the regular values at $s = 2, 3, 4, 5, 6$ in (8.2), respectively. Use $\gamma_0 = -\psi$ and $\psi^{(m)}(x) = (-1)^{m+1}m!\zeta(m+1,x)$ for $m \geq 1$ [6]. To make the

cancellation reproducible, the required intermediate values are

$$\begin{aligned}
\mathcal{H}_2(1+\varepsilon;2,1) &= \frac{1}{720\varepsilon} + \frac{\zeta'(-3)}{96} - \frac{83}{138240} + O(\varepsilon),\\
768\mathcal{H}_2(2;2,1) &= -24\zeta'(-2),\\
768\mathcal{H}_2(3;2,1) &= 24\zeta'(-1) - 4\log 2 + 7,\\
768\mathcal{H}_2(4;2,1) &= -8\zeta'(0) - 10 + 4\log 2,\\
768\mathcal{H}_2(5;2,1) &= -2\gamma - \log 2 + \frac{23}{12},\\
768\mathcal{H}_2(6;2,1) &= \frac{7}{10} - \frac{\pi^2}{15}.
\end{aligned}$$

Finally, $\zeta'(0) = -\frac{1}{2}\log(2\pi)$ and $\zeta'(-2) = -\zeta(3)/(4\pi^2)$ give the printed forms. The latter is the derivative of the zeta functional equation at its simple trivial zero; see also [8]. □

The next Stieltjes coefficient is equally explicit:

$$\boxed{\int_1^3 W(x)\gamma_1(x)\,\mathrm{d}x = \frac{367}{432} - \frac{4}{3}\log 2 - \frac{44}{3}\zeta'(-3) - 4\zeta''(-3).} \tag{8.9}$$

To verify it algebraically, put $\zeta(-3+\varepsilon) = 1/120 + \zeta'(-3)\varepsilon + \zeta''(-3)\varepsilon^2/2 + O(\varepsilon^3)$ and $2^{-\varepsilon} = 1 - \varepsilon\log 2 + \varepsilon^2(\log 2)^2/2 + O(\varepsilon^3)$ in (8.2). The coefficient of $\varepsilon$ is minus $1/768$ times the left side of (8.9); the logarithm-square terms cancel. Theorem 6.1 proves the corresponding finite identity at every higher order.

### 8.1 A convergent hyperbolic integral

At $s = 2$, (5.1) and (3.11) also give

$$\boxed{\int_0^\infty \frac{e^{-2t}\bigl((t^2+3)\sinh t - 3t\cosh t\bigr)}{t^4(1-e^{-t})}\,\mathrm{d}t = \frac{3\zeta(3)}{8\pi^2}.} \tag{8.10}$$

The numerator in parentheses is $t^5/15 + O(t^7)$ at zero, so the integrand has a finite limit. At infinity it decays exponentially with a rational power of $t$. There is no hidden finite part. Multiplying the value of $\mathcal{H}_2(2;2,1)$ by 48 proves the identity.

These examples show two uses of the same kernel: its normal-form coefficients determine a Gamma differential sector, while its beta representation evaluates an independent family of ordinary integrals. They should not be conflated with a claim that all mixed Gamma moments have thereby been evaluated.

## 9 Bernoulli selectors: single-zeta closure and a classification

The beta weight in (5.2) is even and vanishes with its first derivative at both endpoints. A different basis of precisely this polynomial space makes the arithmetic of the interval $[1,3]$ diagonal. Define, for $r \ge 2$,

$$P_r(v) = B_{2r}\left(\frac{v+1}{2}\right) - B_{2r}, \qquad I_r(s) = \int_1^3 P_r(x-2)\zeta(s,x)\,\mathrm{d}x. \tag{9.1}$$

Here $B_n(x)$ is defined by $te^{xt}/(e^t-1) = \sum_{n\ge 0} B_n(x)t^n/n!$, with $B_n = B_n(0)$ [10]. In particular $B_1 = -1/2$ but $B_1(1) = 1/2$. The endpoint derivatives are

$$d_{rj} := P_r^{(j)}(1) = \frac{(2r)!}{2^j(2r-j)!}B_{2r-j}(1), \qquad 1 \le j \le 2r. \tag{9.2}$$

The only nonzero odd derivative in this range is $j = 2r-1$.

## 9.1 The full complex-order selector identity

**Theorem 9.1** (One shifted zeta at every complex order)**.** *As meromorphic functions of $s$,*

$$\boxed{I_r(s) = -\frac{(2r)!}{2^{2r-1}(1-s)_{2r}}\zeta(s-2r) - \sum_{j=2}^{2r}\frac{(-1)^j d_{rj}}{(1-s)_{j+1}}(1+2^{j+1-s}).} \tag{9.3}$$

*The compact integral has only the pole $s = 1$, of residue $-2B_{2r}$; all other apparent poles in the display cancel.*

*Proof.* The Bernoulli reflection identity makes $P_r$ even. The endpoint values and first derivatives vanish. Repeated integration by parts, exactly as in Theorem 5.1, gives

$$I_r(s) = \sum_{j=2}^{2r}\frac{(-1)^j d_{rj}}{(1-s)_{j+1}}\{\zeta(s-j-1,3) - (-1)^j\zeta(s-j-1,1)\}.$$

Integer translation of Hurwitz zeta makes the braces

$$(1-(-1)^j)\zeta(s-j-1) - 1 - 2^{j+1-s}.$$

All even $j$ therefore contribute only elementary terms. For odd $j$, (9.2) vanishes except when $j = 2r-1$, because all odd Bernoulli numbers beyond $B_1$ are zero. At this last index, $d_{r,2r-1} = (2r)!/2^{2r}$. This gives (9.3). Finally, $\int_{-1}^{1} P_r(v)\,\mathrm{d}v = -2B_{2r}$, since the integral of $B_{2r}(x)$ over $[0,1]$ is zero. The compact representation proves the pole assertion. □

**Theorem 9.2** (The selector basis and its uniqueness)**.** *Let $\mathscr{V}_R$ be the rational vector space of even polynomials $P(v)$ of degree at most $2R$ such that $P(1) = P'(1) = 0$, where $R \ge 2$. Then $P_2, \dots, P_R$ form a basis, with the explicit expansion*

$$\boxed{P(v) = \sum_{r=2}^{R}\frac{2^{2r}P^{(2r-1)}(1)}{(2r)!}P_r(v).} \tag{9.4}$$

*Moreover, for a nonzero $P \in \mathscr{V}_R$ and a fixed $r \in \{2,\dots,R\}$, the identity*

$$\int_1^3 P(x-2)\zeta(s,x)\,\mathrm{d}x = F(s)\zeta(s-2r) + G(s) + 2^{-s}H(s) \tag{9.5}$$

*with rational functions $F, G, H \in \mathbb{C}(s)$ holds if and only if $P$ is a nonzero scalar multiple of $P_r$. For rational $P$, that scalar is rational. This is a statement about meromorphic functions of $s$, not about independence of particular numerical zeta values.*

*Proof.* The dimension of the even-polynomial space is $R+1$. Evaluation of $P(1)$ and $P'(1)$ imposes two independent linear conditions, so $\dim \mathscr{V}_R = R-1$. Each $P_r$ belongs to this space and has degree $2r$ with nonzero leading coefficient. They are therefore a basis. For the odd derivative of order $2j-1$ at 1, all basis elements vanish except $P_j$, whose value is $(2j)!/2^{2j}$. This proves (9.4).

It remains to justify genuine functional uniqueness, rather than only uniqueness in a chosen formal list. The functions

$$1, \quad 2^{-s}, \quad \zeta(s-4), \quad \zeta(s-6), \dots, \zeta(s-2R) \tag{9.6}$$

are linearly independent over $\mathbb{C}(s)$. To see this, clear rational denominators in a purported relation. In a right half-plane its Dirichlet expansion has the form

$$\sum_{n\geq 1} Q_n(s) n^{-s} = 0,$$

where each $Q_n$ is a polynomial of uniformly bounded degree in $s$. For $n \geq 3$, it is $Q_n(s) = \sum_{r=2}^{R} p_r(s) n^{2r}$ for fixed polynomials $p_r$; only $n = 1, 2$ have the extra polynomial terms.

If some $Q_n$ were nonzero, choose the smallest such $n_0$ and divide by $n_0^{-s}$ for real $s \to +\infty$. The remaining tail is bounded by a polynomial in $s$ times an exponentially decaying function. This follows from $|Q_n(s)| \leq C(1+s)^M n^{2R}$ and the convergent series $\sum_{n>n_0} n^{2R}(n_0/n)^s$; splitting off half the exponent bounds it by $C'(n_0/(n_0+1))^{s/2}$ for large $s$. A nonzero polynomial $Q_{n_0}(s)$ cannot tend to zero exponentially. Thus every $Q_n$ is zero. For infinitely many integers $n \geq 3$, $\sum p_r(s) n^{2r} = 0$ as a polynomial in $s$; regarding it as a polynomial in $n$ gives $p_r = 0$ for every $r$. The two remaining coefficients also vanish. This proves (9.6).

Expand $P$ by (9.4) and apply Theorem 9.1. The independence just proved forces all coefficients except the prescribed $r$ to be zero in (9.5). The converse follows from (9.3). □

The restriction on the elementary remainder in (9.5) is essential. Allowing arbitrary meromorphic coefficients would make such a classification vacuous. Likewise, this functional theorem says nothing about possible unexpected numerical relations after specializing $s$.

### 9.2 Beta–Bernoulli transport

For the particular polynomial $(1-v^2)^r$, equation (5.6) makes the selector expansion completely explicit:

$$(1-v^2)^r = (-1)^r 2^{2r} \sum_{j=2}^{r} \frac{1}{j} \binom{r}{2j-r-1} \left( B_{2j}\left(\frac{v+1}{2}\right) - B_{2j} \right), \tag{9.7}$$

where a binomial coefficient outside $0 \leq 2j-r-1 \leq r$ is zero. Consequently the Gamma-derived Mellin transform on $[1,3]$ satisfies

$$\mathcal{H}_r(s;2,1) = \frac{1}{4r!(r+1)!} \sum_{j=2}^{r} \frac{1}{j} \binom{r}{2j-r-1} I_j(s). \tag{9.8}$$

This links the affine Gamma square kernel directly to the single-zeta selector basis. It also explains why only certain shifted even-order zeta coordinates appear in its integer-endpoint reduction.

### 9.3 One odd zeta in each trigamma integral

Define the rational number

$$R_r = -\sum_{j=2}^{2r} \frac{(-1)^j d_{rj}}{(j-1)!} (H_{j-1} - 1)(1 + 2^{j-1}). \tag{9.9}$$

**Corollary 9.3** (Bernoulli-weighted trigamma integrals)**.** *For every integer $r \geq 2$,*

$$\boxed{\int_1^3 \left(B_{2r}\left(\frac{x-1}{2}\right) - B_{2r}\right)\psi'(x)\,\mathrm{d}x = R_r + \frac{(-1)^r(2r)!}{2^{4r-2}\pi^{2r-2}}\zeta(2r-1).} \tag{9.10}$$

*Proof.* For any even $P$ with $P(\pm 1) = P'(\pm 1) = 0$, write $d_j = P^{(j)}(1)$. In the finite endpoint expression put $s = 2 + \varepsilon$. The reciprocal denominator is

$$\frac{1}{(1-s)_{j+1}} = \frac{1}{(j-1)!\varepsilon}\left(1 + (H_{j-1} - 1)\varepsilon + O(\varepsilon^2)\right).$$

The endpoint numerator has constant $-1 - 2^{j-1}$ and linear coefficient $2\mathbf{1}_{j\ \mathrm{odd}}\zeta'(1-j) + 2^{j-1}\log 2$. All pole terms cancel by analyticity of the compact integral. The coefficient of $\log 2$ is

$$\sum_{j\geq 2} \frac{(-1)^j 2^{j-1} d_j}{(j-1)!} = -P'(-1) = 0.$$

The remaining value is therefore

$$\sum_{j\geq 2} \frac{(-1)^j d_j}{(j-1)!}\left\{2\mathbf{1}_{j\ \mathrm{odd}}\zeta'(1-j) - (H_{j-1} - 1)(1 + 2^{j-1})\right\}.$$

For $P = P_r$ only $j = 2r - 1$ contributes a zeta derivative, and its coefficient is $-r(2r-1)/2^{2r-2}$. Finally use

$$\zeta'(-2n) = \frac{(-1)^n(2n)!}{2(2\pi)^{2n}}\zeta(2n+1) \quad (n \geq 1),$$

obtained by differentiating the functional equation at a trivial zero [8]. This proves (9.10). □

The first rational offsets are

$$R_2 = 0, \quad R_3 = -\frac{1}{40}, \quad R_4 = \frac{3}{70}, \quad R_5 = -\frac{57}{560}, \quad R_6 = \frac{397}{1155}.$$

For example, the $r = 3$ value is $-1/40 - 45\zeta(5)/(64\pi^4)$, and the $r = 4$ value is $3/70 + 315\zeta(7)/(128\pi^6)$. For $r = 2$ the polynomial weight is $W(x)/16$, reproducing (8.4).

### 9.4 All Stieltjes orders in a single negative-odd zeta jet

The complex-order identity proves more than the trigamma specialization. With $E_j$ from (6.3), define

$$L_r(\varepsilon) = \frac{r}{2^{2r-2}}E_{2r-1}(\varepsilon)\zeta(1 - 2r + \varepsilon) + \sum_{j=2}^{2r} \frac{(-1)^j d_{rj}}{j!}E_j(\varepsilon)(1 + 2^j e^{-\varepsilon\log 2}). \tag{9.11}$$

Equation (9.3) is precisely $I_r(1+\varepsilon) = L_r(\varepsilon)/\varepsilon$. In particular $L_r(0) = -2B_{2r}$, and for every $m \geq 0$,

$$\boxed{\int_1^3 P_r(x-2)\gamma_m(x)\,\mathrm{d}x = (-1)^m m![\varepsilon^{m+1}]L_r(\varepsilon).} \tag{9.12}$$

This is a finite formula in the spectral derivatives $\zeta^{(k)}(1-2r)$, $0 \leq k \leq m+1$, rational numbers, generalized harmonic numbers, and powers of $\log 2$. It uses only one negative-odd spectral center, not an unspecified collection of Stieltjes constants. The coefficient rule (6.4) makes the formula executable at every order. No numerical independence assertion is needed for any of these identities.

# 10 Source audit, verification, and integration status

## 10.1 What was inspected and what was not

The GitHub metadata anchor used for the final source inventory is

`92ee3aaf3087c0e0c29b9ed952e777de5d6accd0`.

The relevant canonical chapters, the immediate predecessor's Gamma sector, and its research-question section were read as TeX through the repository connection. The source manifest in the accompanying package records exact paths and the returned Git blob identifiers. Some initial reads used the default branch; the predecessor's research questions and Gamma-section identity were subsequently checked at the displayed anchor. A Git blob identifier is not being presented as a SHA-256 hash of a locally downloaded file.

The current incoming directory contained five ZIP archives, concerning alternating tails, finite-difference determinants, higher quadratic jets, higher spectral defects, and quartic spectral reconstruction. The repository connection exposed their metadata but not their binary contents. The ZIPs were *not* extracted in this continuation. Matching-title Library TeX abstracts were retrieved for the finite-difference, higher-quadratic-jet, and higher-spectral-defect reports. Their identity with the repository ZIP bytes was not established. These abstracts concern determinantal or spectral factorization problems; they are contextual comparisons, not audited proof dependencies. No assertion that every incoming result has been read, or that the entire repository has been certified, is made.

The main proof dependencies are therefore unusually explicit: the functional-monomial decomposition, the Gamma Bell exponential, elementary weighted integration by parts, and the standard Hurwitz and Lerch identities. The necessary algebraic decomposition is reproved here. The one concrete predecessor question used to measure progress is question 5 of [2, Section 8], concerning the next Gamma projection. Repository files have not been modified remotely.

## 10.2 Classical inputs and the boundary of the novelty claim

The functional-monomial construction is an established component of the integro-differential-algebra literature [3]; its adoption in the canonical manuscript and predecessor is retained with attribution [1, 2]. The pure derivative-square coefficients, the low-index function $f_0$, and the entire $r = 2$ family are inherited specializations. They should not be relabeled as new theorems during integration.

Polynomially weighted Hurwitz integrals and their relations to Bernoulli polynomials, log Gamma, and polygamma functions have extensive classical antecedents; the papers of Espinosa and Moll explicitly study definite and indefinite integrals of this kind [4, 5]. The differentiation, translation, Mellin, Bernoulli, and functional- equation identities used here are standard [7, 6, 9, 10, 8]. Repeated integration by parts itself is not a new method.

The proposed additions relative to the inspected proof chain are the all-$r$ affine square-sector coefficient theorem, the arbitrary symmetric profile extension, their common beta–Hurwitz transport, and the explicit selector classification as a statement over $\mathbb{C}(s)$. The all-order Stieltjes coefficient rules and anchored primitive formulas are proved consequences in the stated normalization. Particular low-order integral values may already occur in the literature. This report does not establish global historical priority for each formula, nor does it claim a new special-function algebraic-independence result at fixed numerical arguments.

### 10.3 The exact proof-status distinctions

The following status list should accompany any manuscript integration.

**Proved here.** Theorem 1.1 gives every $u_0^a u_1^k u_r^2$ coefficient in the canonical inverse-Gamma normal form. Section 4 gives its general symmetric-profile analogue. The Hurwitz and Lerch transforms, their removable-pole prescriptions, all Stieltjes orders, and all anchored shift primitives are analytic identities on their specified domains. Theorem 9.2 classifies exactly the symmetric, endpoint-vanishing polynomial weights with the specified single-zeta functional closure. Its necessity uses a proof of independence over $\mathbb{C}(s)$, rather than independence of a declared formal list.

**A source question partially resolved.** The next-Gamma-projection question is answered for $u_3^2$ with arbitrary powers of $u_0, u_1$ and no additional $u_2$ factor, and simultaneously for every higher derivative square. Terms such as $u_0^a u_1^k u_2^b u_3^2$ with $b > 0$ lie beyond the quadratic high-derivative sector used here. The complete $\sigma_3$ projection is not claimed.

**Not resolved by this report.** The unrestricted unequal-even-tail optimal-prefix conjecture, the Gaussian $S_6$ target, and the current revised $S_8$ candidate remain separate. The formal Gamma theorem does not evaluate all of its finite-part moments. No proof-assistant formalization, independent human peer review, or certified numerical enclosure is claimed.

### 10.4 Corrections, safeguards, and editorial observations

No false inherited mathematical theorem was established by the targeted audit. Several errors would arise from tempting but invalid extensions, and the package supplies explicit guards against them. First, normal reduction must precede truncation of higher derivatives: $\mathcal{N}(u_0 u_4) = u_2^2$, whereas truncating $u_4$ beforehand gives zero. Second, $\mathcal{N}$ is not multiplicative. Third, weighted symbol reduction requires the Appell primitive property in Lemma 2.2; allowing an arbitrary $x$-dependent primitive would destroy the quotient argument. Fourth, removable poles in endpoint formulas must be canceled jointly, not by assigning independent finite parts to their summands. Finally, $B_1(1) = +1/2$ and $B_1(0) = -1/2$ play different roles in the selector calculation. These are proof obligations and implementation safeguards, not accusations that the predecessor omitted them.

A minor editorial duplication was observed in the inspected manuscript README: the paragraph describing the full-Cayley, 1,032-input first-$X$ family appeared twice. Removing one copy would improve the overview without changing any mathematical claim. That suggestion should be checked against the current README before editing, since the repository is changing rapidly. No correction to a numbered theorem is proposed on the basis of that duplication.

### 10.5 Exact finite verification and numerical diagnostics

The executable checks distinguish four kinds of exact finite assertion.

| Replay | Assertions | Scope |
|---|---|---|
| `check_algebra.py` | 1,255 | Normal forms and primitives through weight 18; independent Bell sectors; all-index formula specializations; kernel coefficient identities; deliberately wrong alternatives rejected. |
| `check_endpoint_algebra.py` | 728 | Beta endpoint derivatives and residues; complete-harmonic coefficients; Bernoulli endpoint conventions and rational offsets. |
| `check_profiles.py` | 75 | Three independent polynomial profiles compared with direct Bell expansion and canonical reduction. |
| `check_selectors.py` | 129 | Selector basis, beta–Bernoulli transport, Stieltjes residues, inverse expansions, and an endpoint-sign guard. |
| Total | 2,187 | Finite exact rational or symbolic assertions. |

The algebra replay compares both sides with independently constructed representations. In particular it checks $A_m = N_m + \mathcal{D}T_m$, not merely the displayed selected coefficient. The profile replay does not invoke the proposed profile formula to create its Bell-polynomial reference. The infinite-family assertions nevertheless rest on the mathematical proofs in the article, not the largest computed index.

The numerical program has four separately runnable parts: 15 kernel, 14 transform, 6 Stieltjes, and 21 Lerch/selector comparisons, totaling 56. At 50 working decimal digits it compares beta integrals with hypergeometric values, compact Hurwitz averages with finite endpoint expressions, Mellin integrals with both, and polygamma/Stieltjes quadratures with their explicit evaluations. It includes a complex spectral argument and rational-endpoint root-of-unity filters. Errors are scaled by $\max(1, |L|, |R|)$; small absolute values therefore do not imply the same number of relative digits.

The largest recorded scaled residual is approximately $2.62 \times 10^{-29}$ in a comparison of two *different quadrature orders* for the first Stieltjes average. Its declared tolerance is $10^{-23}$. The comparison of the finer quadrature with the finite endpoint formula is about $5.49 \times 10^{-43}$, and the explicit quartic $\gamma_1$ evaluation agrees to about $4.22 \times 10^{-40}$ in the same scaled measure. The default tolerance for the other comparisons is $10^{-30}$, with explicit local overrides documented in the program. These are floating-point diagnostics, not interval error bounds.

All scripts, exact coefficient tables, normal-form primitives, individual results, environment information, and a replay entry point are included. The integration fragment preserves unique `ag:` labels, the distinction between ordinary and finite-part integrals, and the nonmultiplicative meaning of the normal projection.

## 11 Further research questions and concrete next targets

The following problems continue the exact-identity programme rather than asking for bounds or extrapolating a numerical pattern. The proved formulas give initial conditions and explicit checks for each.

**1. Complete the coupled next Gamma sector.** Find a finite expression for the coefficient

$$[u_0^a u_1^k u_2^b u_3^2] N_{a+2k+3b+8}$$

for every $b > 0$. The present theorem supplies $b = 0$, while the predecessor supplies the pure $u_2$ projection. A quadratic background for $u_0$ would retain $u_2$ but replace constant-coefficient weighted adjoints by variable- coefficient ones. The precise missing step is a canonical symbol calculus compatible with that background and the original normal basis. A formula should include the parity zeros and the retained endpoint primitive, not just the first few nonzero weights.

**2. Determine the first genuinely nonlinear polarization.** The affine quadratic symbol is a polynomial in one invariant $\eta$. At cubic and quartic perturbation order there are several symmetric invariants after imposing $\sum \xi_i + \kappa = 0$. Identify an invariant basis which maps triangularly to the functional-monomial normal form. Use it to compute complete families with three or four high-derivative factors, distinguishing cubic terms such as $u_2^3$ from quartic terms such as $u_2^2 u_3^2$. The source's all-$u_2$ formulas provide stringent independent checks but not the mixed-sector solution.

**3. Classify elementary symmetric profiles.** For which even analytic profiles $K$ do all square-sector kernels $g_{K,r}$ belong to a fixed finite combination of hyperbolic functions and their divided differences? The finite Gamma mixtures in Section 4 give one class. Seek a converse under explicit growth and analyticity hypotheses, or exhibit another class coming from an interpretable Gamma or Barnes ratio. A claim about minimality must specify the allowed coefficient field and operations.

**4. Generalize the selector classification to other intervals.** On $[1, 3]$, integer translation leaves only the bases 1 and $2^{-s}$ besides shifted Riemann zeta functions. For intervals $[A, B]$ with positive integer endpoints, prescribe the finite exponential remainder and characterize weights selecting exactly one shifted zeta function. For rational endpoints the natural coordinates are Hurwitz values or Dirichlet $L$-functions. The goal is an exact functional classification, including endpoint multiplicities and a proof of the necessary Dirichlet-series independence statement.

**5. Find Lerch selectors with a prescribed polylogarithm order.** The general Lerch endpoint formula is finite, but the rational factors $z^{-A}$ prevent the simple odd-derivative diagonalization used at $z = 1$. Characterize polynomial weights, possibly depending rationally on $z$, whose transform has one polylogarithm order plus an elementary remainder. Determine whether the required endpoint conditions form a finite interpolation problem, and prove its confluence as $z \to 1$. The explicit quartic identity in Section 7 is a normalization test rather than a general classification.

**6. Put all selector Stieltjes orders in a uniform closed form.** Equation (9.12) is a finite coefficient rule at every order. Seek a compact bivariate generating identity in the selector degree and Stieltjes order that preserves the single spectral center and makes the harmonic coefficients explicit. Determine which anchored integrals in the shift can be expressed through a fixed family of multiple-Gamma or zeta-derivative functions. Any such primitive must retain the anchor conditions from Section 6.

**7. Describe the numerical image of the Gamma square sector.** The coefficient theorem is formal. Evaluate or systematically reduce the ordinary shifted integrals and the original-coordinate finite parts of $\psi^a(\psi')^{2k}(\psi^{(r)})^2$ in a declared basis of convergent multiple-zeta derivatives or polylogarithmic integrals. The derivative- ideal pole calculus in the manuscript supplies convergent coordinates, but it does not establish their arithmetic minimality. Retain the primitive endpoint term when transporting the result to inverse-Gamma moments.

**8. Return to the unequal-tail boundary determinant.** The all-$r$ Gamma theorem does not prove the unequal-even-tail optimal-prefix conjecture. The next unresolved general fixed-gap problem in the inspected predecessor is the three-column boundary determinant for gap six. Find an exact scalar recurrence, a determinant identity, or an arithmetic normalization that proves nonvanishing.

Entrywise positivity and pairwise ratio separation are not sufficient. Any resulting inverse row can then be combined with the predecessor's already proved universal finite telescoper.

**9. Build a proof-carrying coefficient compiler.** Formalize the terminating normal-form algorithm and its identity $P = \mathcal{N}(P) + \mathcal{DT}(P)$ in a proof assistant. Add the Appell covariance lemma and the weighted quadratic symbol as reusable algebraic results. A useful initial target is a checker for the supplied rational JSON certificates, followed by the all-order coefficient proof. Certified real or complex interval checks of the displayed analytic integrals would be a separate numerical project, not a replacement for analytic continuation proofs.

**10. Preserve the original period questions.** The Gaussian $S_6$ and current revised $S_8$ candidates call for exact polylogarithmic identities with fixed branches and complete lower-weight constants. Neither selector closure nor normal-form coefficient closure settles them. A promising next step is a parameter identity whose rigorously controlled boundary value matches the precise frozen candidate, rather than a new unconstrained integer-relation search.

### Conclusion

An affine perturbation converts an infinite family of Gamma normal-form coefficients into one entire divided difference. The beta representation of its coefficients supplies a common route to finite Hurwitz and Lerch endpoint formulas, generalized Stieltjes averages, and anchored primitives. Bernoulli endpoint data then diagonalize the integer- interval arithmetic and give a genuine single-zeta functional classification. These are exact identities with separate formal, analytic, and computational meanings. Their limitations identify the next nonlinear and arithmetic questions without weakening the proved all-index statements.

## References

[1] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*. Canonical manuscript and the chapters `07-ibp-normal-forms.tex` and `07-differential-polynomial-moments.tex`. Repository metadata anchor `92ee3aaf3087c0e0c29b9ed952e777de5d6accd0`; consulted 11 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/92ee3aaf3087c0e0c29b9ed952e777de5d6accd0/Analysis/Polylogarithms/docs/manuscript.

[2] *Exact Harmonic-Tail Reconstruction and Variational Gamma Identities*. ProveIt working report, 11 October 2026. Especially Section 4, `sections/04_gamma_sectors.tex`, and Section 8, `sections/08_audit_research.tex`, research question 5. https://github.com/VladimirReshetnikov/ProveIt/tree/92ee3aaf3087c0e0c29b9ed952e777de5d6accd0/Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/harmonic-tail-gamma. The report's internal historical baseline is different from this continuation's metadata anchor; the source manifest records that distinction.

[3] Li Guo, Georg Regensburger, and Markus Rosenkranz, *On integro-differential algebras*. Journal of Pure and Applied Algebra **218** (2014), no. 3, 456–473. https://doi.org/10.1016/j.jpaa.2013.06.015. Preprint: https://arxiv.org/abs/1212.0266. The functional-monomial construction is a classical input.

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[10] NIST Digital Library of Mathematical Functions, §24.2, *Definitions and Generating Functions*, and §24.4, *Basic Properties*. Consulted 11 October 2026. https://dlmf.nist.gov/24.2; https://dlmf.nist.gov/24.4.